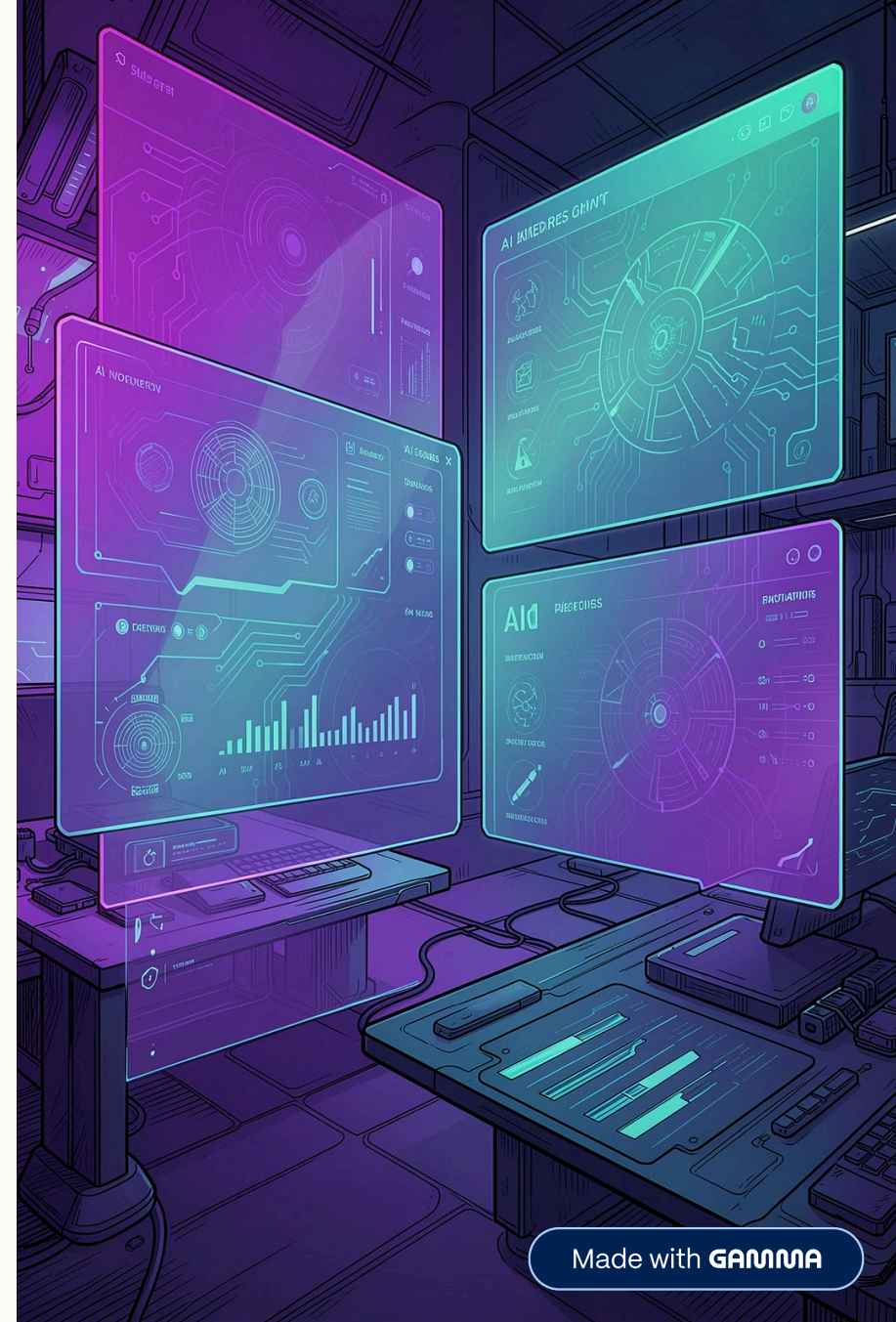


PatentVerse

AI-Powered Multi-Agent Patent Intelligence Platform

Team: Kusuma R · Meghana R · Monikha M · Raksha R

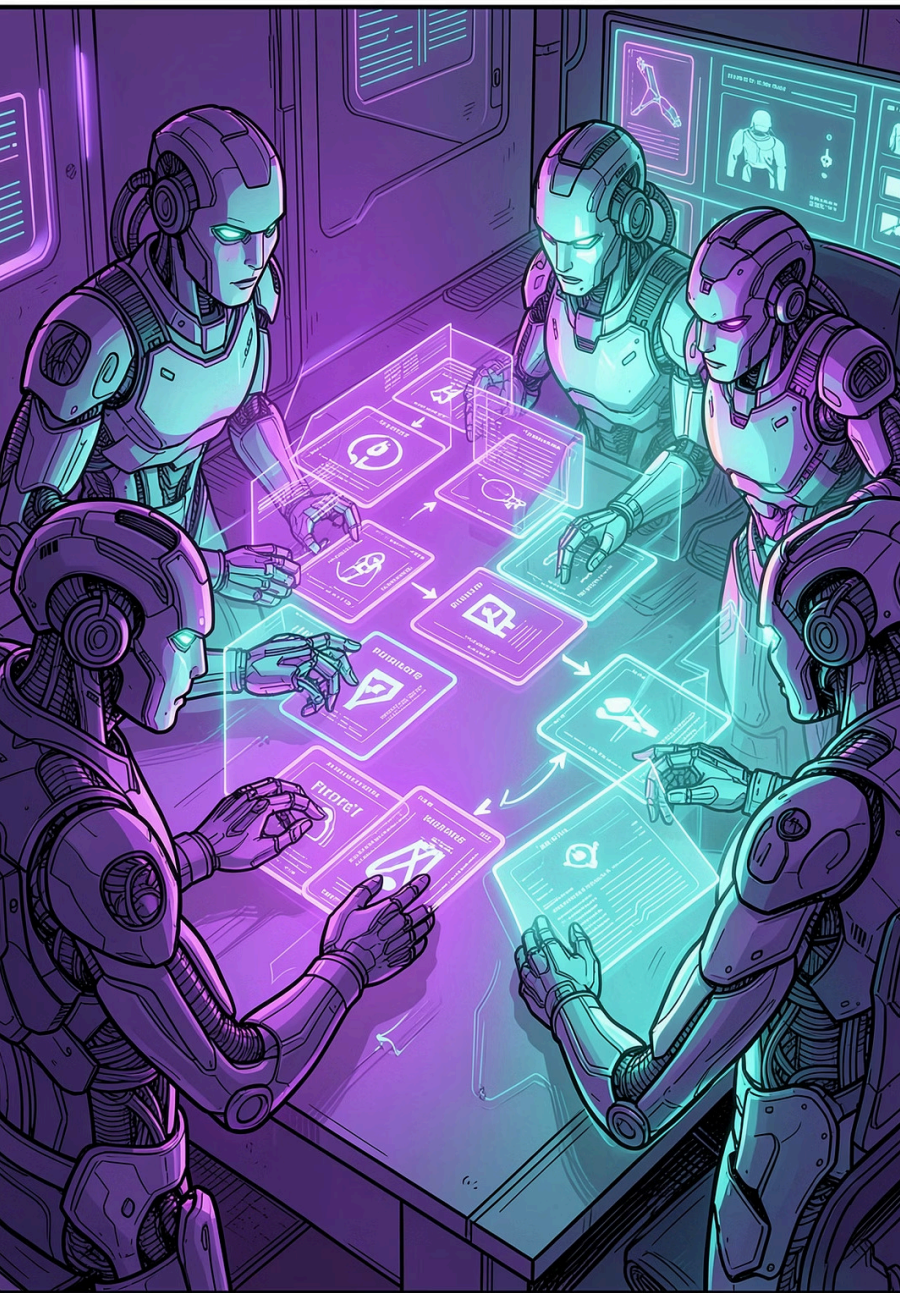


Made with GAMMA

The Problem

- Patent searches are manual, slow, and error-prone.
- Evaluating innovation requires scarce domain expertise.
- Existing systems lack proactive AI assistance and synthesis.
- Assessing legal, technical, and market risk is fragmented and opaque.





Our Solution — PatentVerse

An enterprise-grade platform that coordinates specialized AI agents to discover prior art, generate innovation concepts, analyze risks, and deliver a concise AI verdict — at scale.



Search

Automated similar-patent discovery across global sources.



Generate

AI-driven feature and product innovation suggestions.



Analyze

Risk scoring: legal, technical, and market dimensions.



Verdict

Concise AI verdict summarizing patentability and action items.

Multi-Agent Architecture



Each agent runs specialized pipelines and shares structured signals to produce a single, actionable verdict.

Patent Agent

Similar-Patent Discovery

Semantic and citation-aware retrieval across global patent corpora.

Prior-Art Identification

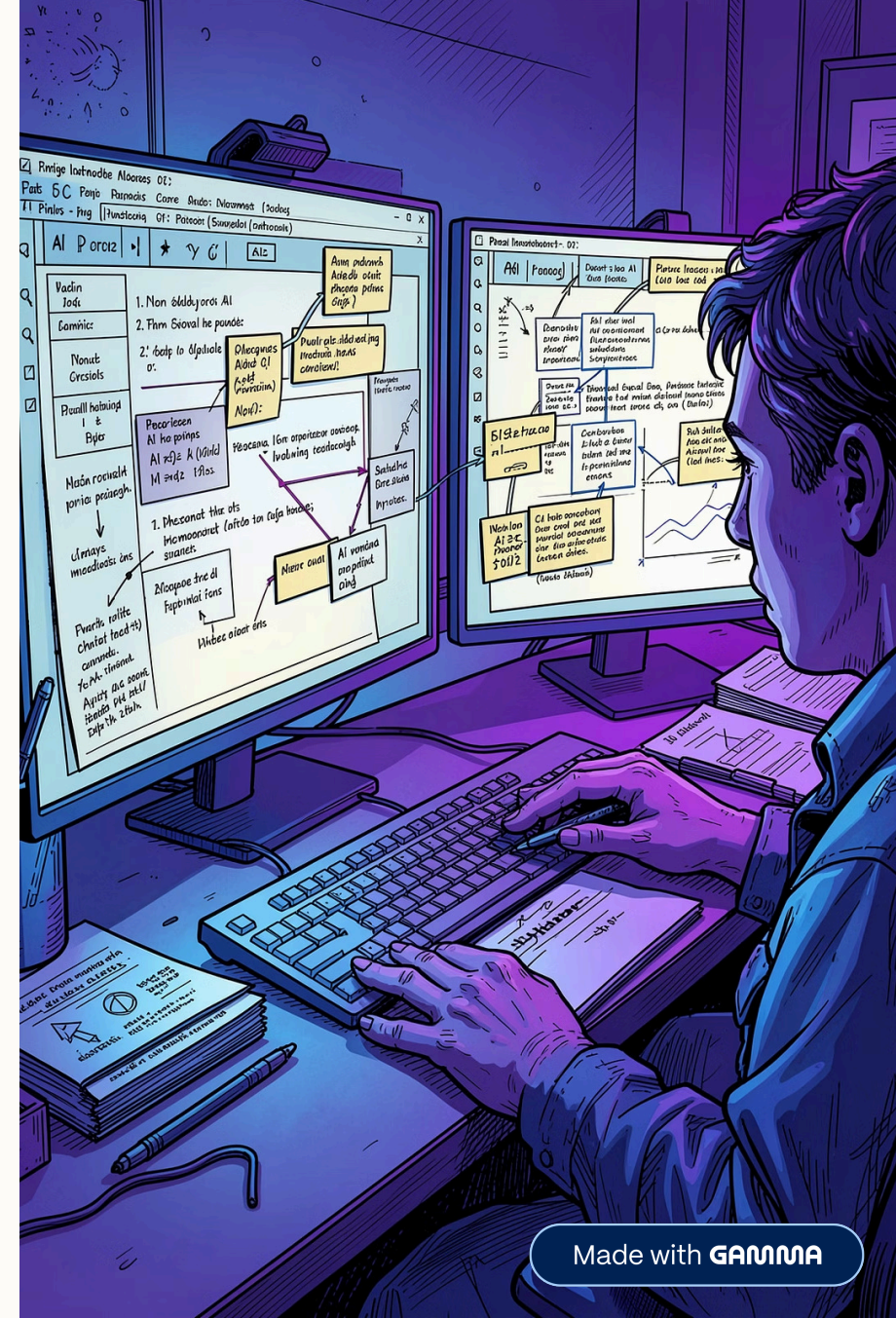
Ranked prior-art evidence with source provenance.

Novelty Analysis

Automated novelty checks with claim-level comparisons.

Patent Risk Estimation

Likelihood indicators for infringement and rejection risk.



Innovation Agent

Feature Suggestions

Context-aware, patent-aware feature concepts tailored to product goals.

Product Improvements

Engineering-ready improvement blueprints with complexity estimates.

Commercial Opportunities

Target markets and commercialization vectors prioritized by ROI signals.

Future Enhancements

Scalable feature roadmaps aligned with IP strategy.

Risk Agent — Scoring & Indicators

1

Patent Risk

Claim overlap, prior-art density,
litigation exposure score.

2

Market Risk

Adoption likelihood, competitor
saturation, TAM signal.

3

Technical Risk

Feasibility, prototype readiness,
integration complexity.

Risk cards combine normalized scores (0–100) and color-coded indicators to guide prioritization.

Technology Stack



Frontend

Streamlit — rapid prototyping, secure enterprise UI



Backend

Python microservices, scalable APIs, async workers



AI

Google Gemini API for multi-agent orchestration and LLM reasoning



Tools

GitHub, VS Code — CI/CD, code review, and reproducible pipelines

Dashboard — Results at a Glance

Key result tiles displayed in one view:

- **Innovation Score** — composite of novelty, commercial fit, and technical readiness.
- **Market Potential** — opportunity heatmap and TAM estimate.
- **Patent Risk** — normalized legal risk gauge with sources.
- **AI Verdict** — one-line recommendation plus next actions.

82

Innovation
Score (0–100)

64

Market
Potential

28

Patent Risk
Lower is better

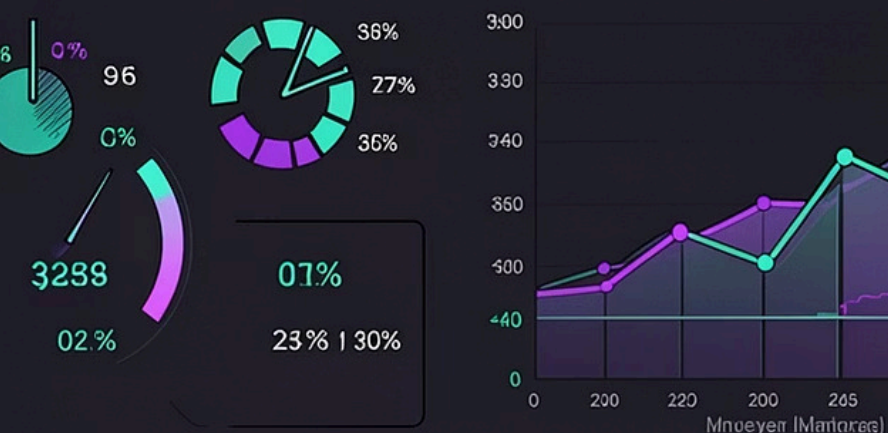
Approve

AI Verdict
Priority recommendation

AI Dashboard

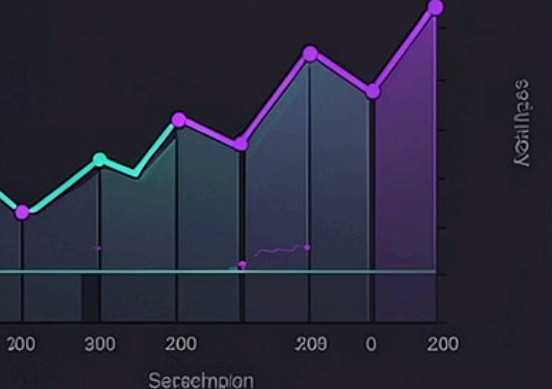
Henry Sep 15

Innovation Score Gauge



Innovation Score

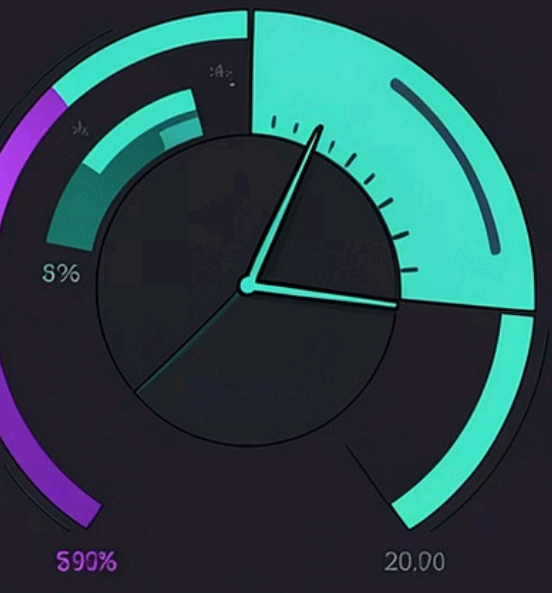
Market Potential



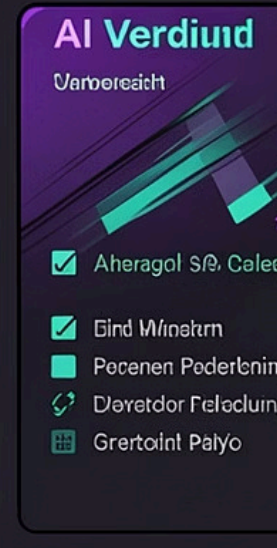
Patent Risk

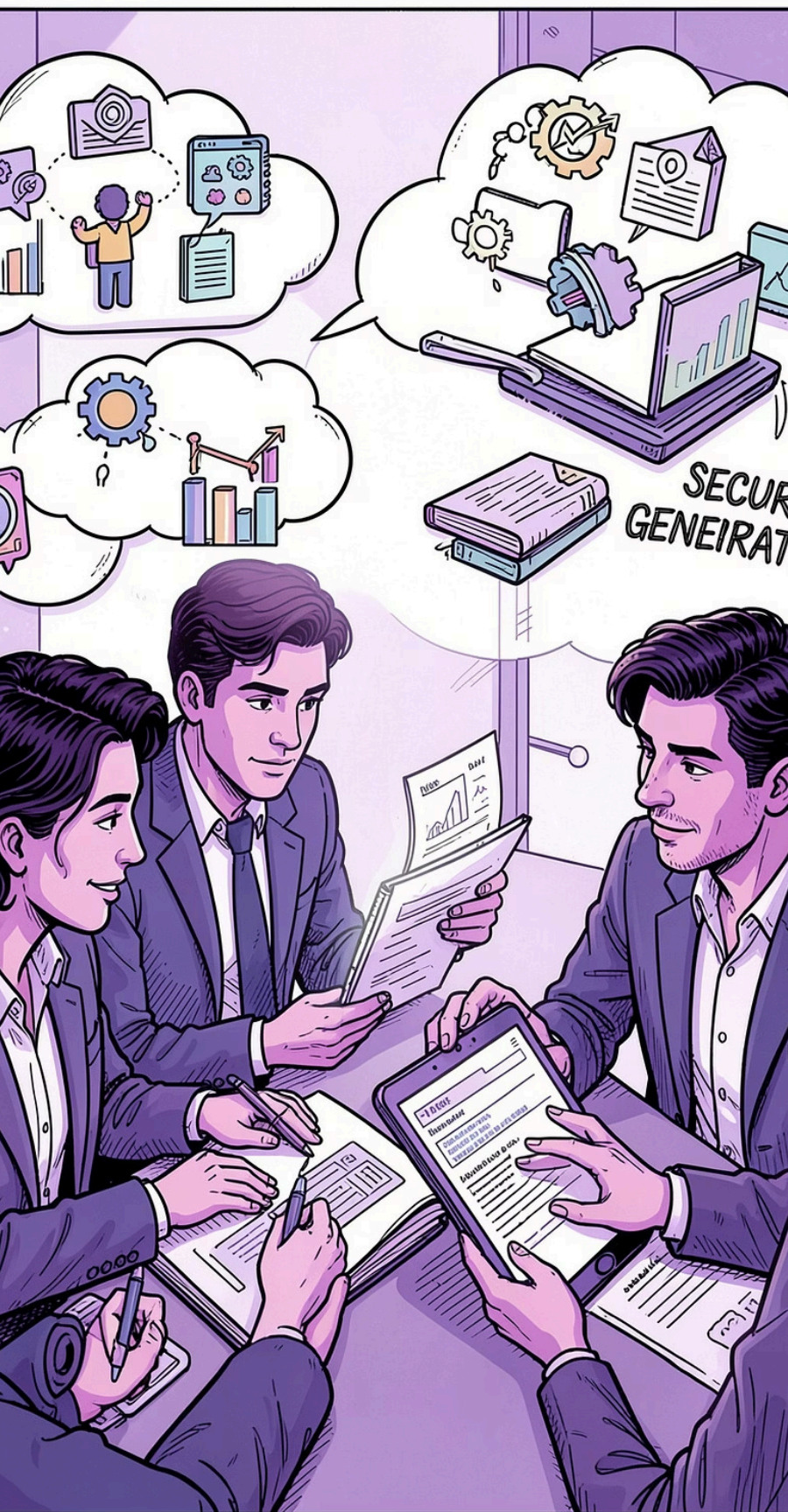


Patent Risk



AI Verdict





Conclusion & Future Scope

PatentVerse accelerates patent analysis, generates actionable innovation, and centralizes multi-dimensional risk evaluation with a single AI verdict for go/no-go decisions.



Integrations

Planned connectors to USPTO, EPO, and commercial patent databases.



Cloud & Scale

Containerized deployment, multi-tenant SaaS, and secure enterprise SSO.



Reporting

Executive PDFs, exportable evidence stacks, and audit-ready provenance.



Roadmap

Expand agent suite, continuous learning from legal outcomes, and partnership integrations.



Investor-ready: clear monetization paths (SaaS tiers, enterprise integrations, professional services) and defensible IP tooling.